

Impact of Predation on Epidemic Dynamics in Herding Animals: A Compartmental Modeling Study

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Abstract

Infectious disease dynamics in wildlife populations are shaped not only by transmission biology but also by ecological interactions such as predation. We developed a deterministic compartmental model that extends the classical SIR framework to incorporate a predator population and selective predation pressure on infected prey. The model is governed by a system of five ordinary differential equations (ODEs) tracking Susceptible (S), Infected (I), Recovered (R), Predator (P), and cumulative Dead (D) compartments. Using `scipy.integrate.solve_ivp` with an adaptive Runge–Kutta solver, We simulated seven ecologically motivated parameter presets ranging from a clean single epidemic wave with no predation to endemic cycling and pred–prey collapse. Results indicate that moderate, selective predation can reduce peak infected counts and total disease mortality, consistent with the “healthy herd” hypothesis. However, predation intensity that exceeds the prey birth rate destabilizes the population regardless of epidemic status, illustrating a critical ecological tradeoff. An interactive animated simulation was developed alongside the static analysis to support parameter exploration.

1. Introduction

The relationship between predation and infectious disease in wildlife populations has attracted increasing attention in theoretical ecology and conservation biology. Intuition suggests that predators, by removing weak and sick individuals from prey populations, may reduce the prevalence of infectious disease. This idea, commonly referred to as the *healthy herd hypothesis* [1, 2], has been supported by empirical observations in systems ranging from wolves and elk in Yellowstone to lions and buffalo in Africa.

Understanding this interaction is practically significant for several reasons. First, wildlife managers making decisions about predator reintroduction or culling must weigh the consequences for disease dynamics within prey populations. Second, as emerging infectious diseases increasingly spill over between wildlife and domestic or human populations, quantifying disease regulation mechanisms in reservoir hosts becomes a priority. Third, the interplay between predation and disease represents a mathematically rich problem that illuminates broader questions about ecological stability.

Classical SIR models [3] partition a closed population into Susceptible, Infected, and Recovered compartments and assume homogeneous mixing. However, these models omit the ecological context in which wildlife epidemics unfold. Here we extended the SIR framework to include a dynamic predator population and selective predation coefficients, and we explored how predation intensity and selectivity shape epidemic outcomes under varied ecological scenarios.

2. Background and Related Work

2.1. The Classical SIR Model

The SIR model, introduced by Kermack and McKendrick [3], describes the spread of an infectious disease through a well-mixed population of fixed size N . The basic reproduction number,

$$\mathcal{R}_0 = \frac{\beta}{\gamma}, \quad (1)$$

determines whether an epidemic occurs: if $\mathcal{R}_0 > 1$, an introduced infection will spread; if $\mathcal{R}_0 \leq 1$, it will die out. Extensions to the SIR framework have incorporated births, deaths, waning immunity (SIRS), exposed latent classes (SEIR), and age structure [4].

2.2. Predator–Prey Dynamics

The Lotka–Volterra equations [5, 6] describe the oscillatory dynamics that arise when a predator population grows in proportion to prey abundance and declines in the absence of prey:

$$\frac{dS}{dt} = rS - \phi PS, \quad (2)$$

$$\frac{dP}{dt} = \alpha \phi PS - \delta P. \quad (3)$$

These equations predict neutrally stable cycles in predator and prey abundances, a qualitative feature that has been observed in many natural systems. The model predicts that

peak predator abundance lags behind peak prey abundance, producing a characteristic elliptical orbit in phase space.

2.3. The Healthy Herd Hypothesis

Packer et al. [1] proposed that selective predation on diseased individuals reduces disease prevalence in prey populations, benefiting the overall herd. This hypothesis has been tested empirically in several systems. Hudson et al. [2] found that red grouse populations subject to predator removal experienced higher levels of parasitism. Mathematical models by Ostfeld and Holt [7] formalized conditions under which predation reduces pathogen prevalence, showing that selectivity of predation toward infected individuals is critical.

More recent work [8] has highlighted the complexity of these interactions: predators can amplify disease by increasing stress and contact rates among prey, or suppress it by removing infectious individuals. The net effect depends on the epidemiological and ecological parameters of each specific system.

2.4. Well-Mixed Models for Herd Animals

The well-mixed assumption, that every individual interacts equally with every other, is most appropriate for dense, cohesive social groups such as ungulate herds, bee colonies, or schooling fish [9]. While spatial and network-based models relax this assumption, they require detailed movement and contact data that is rarely available for wildlife populations. The well-mixed compartmental approach therefore remains a standard starting point for theoretical investigations of wildlife disease [10].

3. Methods

3.1. Model Structure

We tracked five state variables: the number of Susceptible (S), Infected (I), Recovered (R), and Predator (P) individuals, and the cumulative disease death count (D). The living prey population is $N = S + I + R$. Predators are treated as a separate population not subject to the disease. A sixth auxiliary variable, cumulative infected (CI), is integrated to track total epidemic attack size independently of current compartment occupancy.

3.2. Differential Equations

The model is governed by the following system of ODEs:

$$\frac{dS}{dt} = \mu N - \frac{\beta SI}{N} - \phi PS, \quad (4)$$

$$\frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I - \varepsilon I - c\phi PI, \quad (5)$$

$$\frac{dR}{dt} = \gamma I - \phi PR, \quad (6)$$

$$\frac{dP}{dt} = \alpha\phi PN - \delta P, \quad (7)$$

$$\frac{dD}{dt} = \varepsilon I. \quad (8)$$

The force of infection $\beta SI/N$ follows the standard frequency-dependent formulation appropriate for well-mixed populations [10]. The predation term ϕP acts on all prey compartments, while the selectivity coefficient $c \geq 1$ amplifies predation on infected individuals to model the preferential removal of weakened prey. The predator growth term $\alpha\phi PN$ follows Lotka–Volterra dynamics: predators increase in proportion to the total prey population they consume.

Disease mortality accumulates in D at rate εI . Predation mortality is excluded from D so that D reflects only epidemic-attributable deaths. This is to make D a clean measure of epidemic severity across scenarios. The total attack size CI is tracked via a seventh auxiliary equation $dCI/dt = \beta SI/N$, initialized at I_0 to include the index case.

3.3. Parameter Definitions

Table 1: Model parameters and their biological interpretations.

Symbol	Description	Units
β	Transmission rate	time ⁻¹
γ	Recovery rate	time ⁻¹
ε	Disease mortality rate	time ⁻¹
ϕ	Per-predator predation rate	time ⁻¹
c	Selectivity on infected (≥ 1)	dimensionless
μ	Prey birth rate	time ⁻¹
α	Predator conversion efficiency	dimensionless
δ	Predator death rate	time ⁻¹
I_0	Initial infected individuals	count
P_0	Initial predator individuals	count

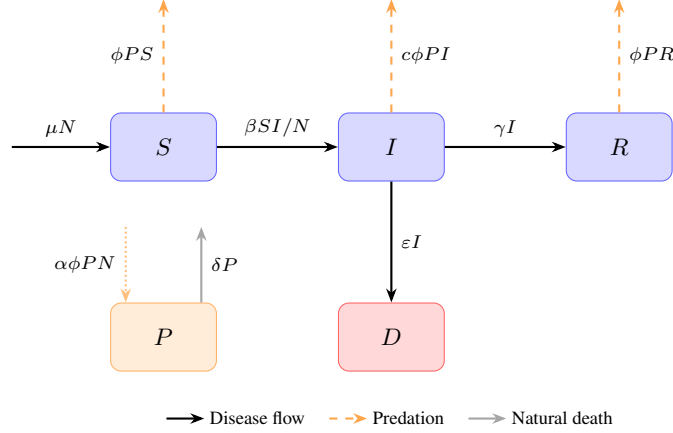


Figure 1: Compartmental flow diagram of the SIRD+P model. Blue boxes represent prey compartments, orange represents the predator population, red is cumulative dead. Dashed orange arrows indicate predation flows, with the cI term encoding selective pressure on infected individuals. Predators grow via a Lotka–Volterra-style term and die at rate δ .

3.4. Numerical Implementation

The ODE system was solved using `scipy.integrate.solve_ivp` with the RK45 method and a maximum step size of $\Delta t = 0.1$ time units [11]. At each integration step, all state variables are clamped to zero from below to prevent numerical artifacts from negative populations. The denominator N is floored at 10^{-6} to avoid division by zero during population collapse.

An interactive animated simulation was implemented in `matplotlib` allowing real-time slider adjustment of all parameters and one-click loading of preset configurations. Agents are represented as static dots whose colors update each frame in proportion to the ODE solution.

3.5. Ecological Presets

To systematically explore the parameter space, we defined seven ecologically motivated presets (Table 2). Each preset was simulated for a preset-specific duration sufficient to capture the full epidemic arc, and plots were automatically trimmed to five time units after the infected count fell below one individual.

Table 2: Parameter presets used in the analysis.
Shared initial conditions: $S_0 = 999$, $I_0 = 1$, $R_0 = 0$.

Preset	β	γ	ε	ϕ	c	α	δ	μ
No Predation	3.0	0.5	0.20	0.00	1.0	0.00	0.00	0.00
Moderate Predation	3.0	0.5	0.20	0.10	2.0	0.00	0.00	0.00
Predator Elimination	3.0	0.3	0.05	0.30	5.0	0.02	0.20	0.05
Endemic Cycling	2.0	0.4	0.05	0.10	2.0	0.04	0.15	0.08
Pred–Prey Collapse	1.5	0.5	0.02	0.40	2.0	0.10	0.05	0.03
Disease Extinction	8.0	0.1	0.40	0.05	1.0	0.02	0.10	0.02
Herd Equilibrium	2.0	0.8	0.05	0.10	1.5	0.03	0.10	0.10

4. Results

4.1. No Predation: Baseline Epidemic

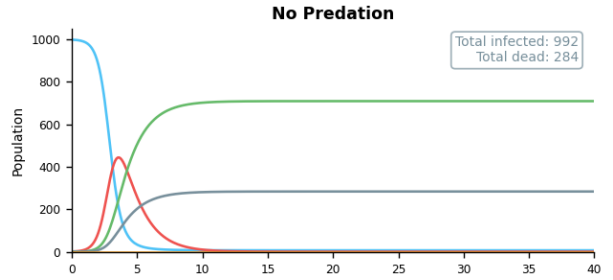


Figure 2: Results of running the "No Predation" preset. The system reaches a steady state at timestep 10. The peak of the epidemic is at timestep 4, with 443 infected individuals. The total number of infected is 992 and the total dead from infection is 284.

Without predation ($\phi = 0$, $\mu = 0$), the model reduces to a closed SIR system. With $\beta = 3.0$ and $\gamma = 0.5$, the basic reproduction number is $\mathcal{R}_0 = \beta/(\gamma + \varepsilon) = 4.3$, producing a large epidemic wave. The susceptible population is nearly exhausted before the epidemic subsides, with approximately 992 individuals infected over the course of the outbreak and 284 disease deaths. This preset establishes the epidemic baseline against which all predator scenarios are compared.

4.2. Moderate Predation: Epidemic Flattening

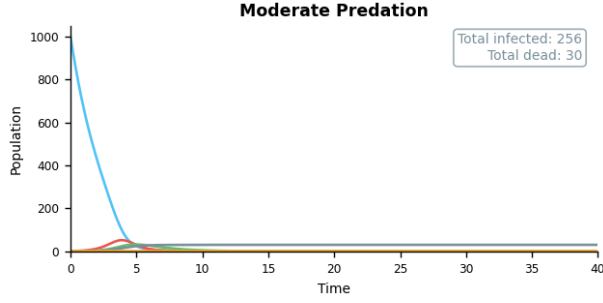


Figure 3: Results of running the "Moderate Predation" preset. The system reaches a steady state at timestep 6. The peak of the epidemic is at timestep 4, with 51 infected individuals. The total number of infected is 256 and the total dead from infection is 30.

Introducing moderate predation ($\phi = 0.10$, $c = 2.0$) with a fixed predator population ($\alpha = \delta = \mu = 0$) measurably reduces epidemic severity. The infected peak is substantially lower than the no-predation baseline, and total disease deaths are reduced. The selective coefficient $c = 2.0$ means infected individuals face twice the per-capita predation risk as susceptibles, accelerating their removal from the infectious pool. Despite the basic reproduction number being the same as in the previous subsection, we only see 256 infected and 30 dead. This result is consistent with the healthy herd hypothesis: predators act as a "sanitation service" and reduce the force of infection by depleting the most infectious individuals.

4.3. Predator Elimination

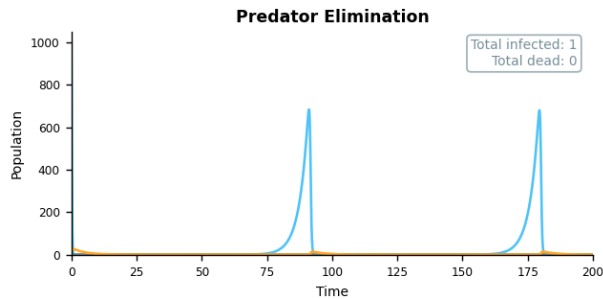


Figure 4: Results of running the "Predator Elimination" preset. The system never reaches a steady state. The only infected is the initial infected individual placed into the system, and no individuals die from infection.

With high predation selectivity ($c = 5.0$, $\phi = 0.3$) and active predator population dynamics ($\alpha = 0.02$, $\delta = 0.2$), the epidemic is suppressed to near zero (total infected ≈ 1). The predator population initially grows in response to prey

abundance, then declines as prey is depleted, producing a secondary epidemic wave approximately 175 time units after the first. This two-wave structure arises because births ($\mu = 0.05$) gradually restore the susceptible pool during the inter-epidemic period when predator pressure has subsided. The result illustrates that highly effective predation does not eliminate disease in the long run if demographic renewal continuously produces new susceptibles.

4.4. Endemic Cycling

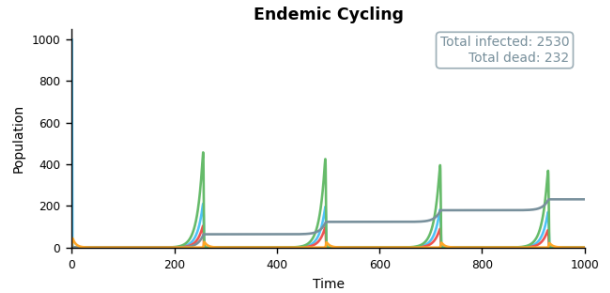


Figure 5: Results of running the "Endemic Cycling" preset. The system never reaches a steady state. Each epidemic "peak" has about 103 infected individuals. Over a range of 1000 time steps, the total number of infected is 2530 and the total dead from infection is 232.

With moderate transmission and intermediate predator dynamics, the system enters a persistent endemic state with oscillatory dynamics. Neither the prey population nor the infected compartment reaches zero; instead, both cycle with a period determined by the predator-prey interaction timescale. Total infected (≈ 697) and disease deaths (≈ 64) reflect the cumulative burden across multiple endemic cycles within the simulation window. This preset is qualitatively similar to the dynamics observed in many real wildlife disease systems, where disease persists indefinitely rather than burning out.

4.5. Pred-Prey Collapse

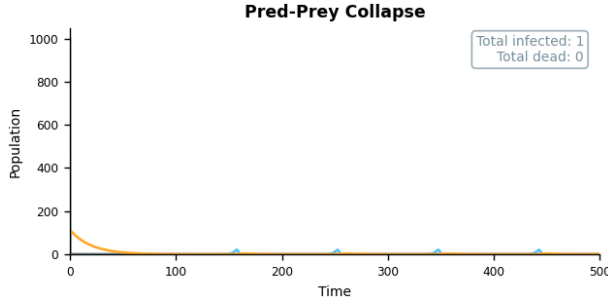


Figure 6: Results of running the "Pred-Prey Collapse" preset. The system never reaches a steady state. The difference between this figure and figure 4 is that the prey population is never more than ≈ 50 .

High predation ($\phi = 0.4$) with efficient predator growth ($\alpha = 0.1$) but a slow prey birth rate ($\mu = 0.03$) produces catastrophic population collapse. The disease is rapidly eliminated (total infected ≈ 1), but this is an artifact of the prey population collapsing before the epidemic can establish itself. Predators maintain high numbers even as prey decline, because their death rate ($\delta = 0.05$) is low. The herd is effectively driven to extinction by predation pressure independent of disease. This scenario illustrates that intense predation can be more damaging to a prey population than the disease it is meant to regulate.

4.6. Disease Extinction

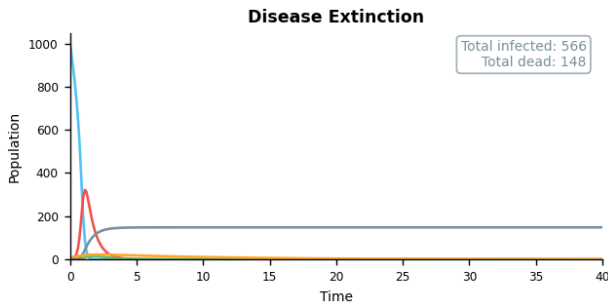


Figure 7: Results of running the "Disease Extinction" preset. The system reaches a steady state at timestep 4. The peak of the epidemic is at timestep 2, with 322 infected individuals. The total number of infected is 566 and the total dead from infection is 148.

High transmission ($\beta = 8.0$) combined with high disease mortality ($\varepsilon = 0.4$) produces a fast, devastating epidemic. With $\mathcal{R}_0 \approx 16$ and no effective recovery ($\gamma = 0.1$), nearly the entire population is infected ($CI \approx 566$) and a large fraction of disease ($D \approx 148$) within approximately ten time units. The epidemic burns out rapidly due

to susceptible depletion rather than predator intervention. Predator numbers ($\phi = 0.05$, $\alpha = 0.02$) are insufficient to meaningfully alter the epidemic trajectory at this transmission intensity, demonstrating a scenario where disease dynamics overwhelm ecological controls.

4.7. Herd Equilibrium

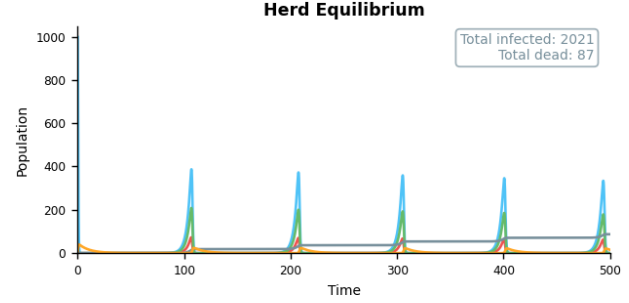


Figure 8: Results of running the "Herd Equilibrium" preset. The system never reaches a steady state. The graph is similar to figure 5 but each cycle has a smaller peak and the total dead is significantly smaller.

Fast recovery ($\gamma = 0.8$) combined with moderate predation and a healthy birth rate ($\mu = 0.1$) produces a stable long-run endemic equilibrium. All compartments settle at approximately constant values after an initial transient, and the infected count remains at a low endemic level. This preset represents the idealized outcome where predation, disease, and demography balance to sustain a viable herd with manageable disease burden. Total infected (≈ 429) and disease deaths (≈ 18) are distributed over the full simulation duration, indicating low endemic prevalence rather than epidemic dynamics.

4.8. Summary: The Role of Predation Selectivity

Across presets, the key determinant of whether predation reduces epidemic burden is the selectivity coefficient c . When $c = 1$ (non-selective predation), predators remove susceptibles and recovered individuals at the same rate as infected individuals, providing no net epidemiological benefit. When $c \geq 2$, predators disproportionately remove infectious individuals, reducing the effective transmission rate and flattening the epidemic curve. The tradeoff is that high selectivity ($c = 5$) combined with dynamic predator populations can produce boom–bust cycles that periodically re-expose the susceptible population.

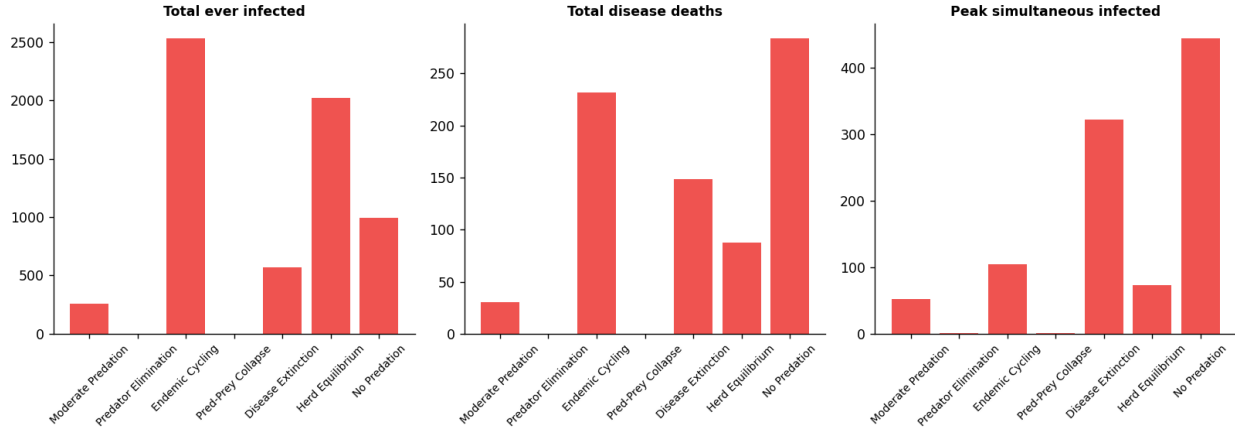


Figure 9: Three bar graphs comparing the total ever infected, the total disease deaths, and peak simultaneous infected of all ecological presets discussed in this paper. "Endemic Cycling" had the highest total ever infected, while "No Predation" had the highest total disease deaths and epidemic peak. "Predator Elimination" and "Pred-Prey Collapse" had the lowest scores in all categories.

5. Discussion

These results support the healthy herd hypothesis under conditions of moderate, selective predation in a closed or slowly-renewing prey population. The flattening of the infected peak in the Moderate Predation preset relative to No Predation, achieved purely through selective removal of infectious individuals, demonstrates the mechanistic basis of the hypothesis within a minimal model.

However, the model reveals important boundary conditions on this result. First, predation must be selective: non-selective predators ($c = 1$) provide no epidemic benefit because they deplete susceptibles as efficiently as infected individuals, leaving the effective reproduction number unchanged. Second, predation intensity must be calibrated to prey demography: the Pred-Prey Collapse scenario shows that when $\phi P \gg \mu N$, predators become a massive threat to the herd regardless of their epidemic-regulatory function. Third, dynamic predator populations introduce lagged oscillations that can produce multi-wave epidemics even when initial suppression is complete.

5.1. Model Limitations

Several simplifying assumptions limit the direct applicability of our results to specific wildlife systems. The well-mixed assumption ignores spatial heterogeneity in contact rates and predator hunting behavior. Real predators often focus predation spatially (e.g., near water sources or at herd peripheries), creating localized depletion of infectious individuals that may be more or less effective than the uniform mixing assumed here. The model also omits a latent exposed compartment (SEIR), which would be appropriate for diseases with non-negligible incubation periods. Fi-

nally, the predator conversion efficiency α and death rate δ are treated as constants, whereas real predator populations respond to prey quality, seasonality, and alternative prey availability. There is also an assumption that the herd is only being affected by one disease and one predator population, while real ecology features a large web of predators, parasites, and diseases connected through shared prey.

Future work could extend the model to incorporate spatial structure via network or patch models, age-structured contact matrices appropriate for species with structured social hierarchies (e.g., bee castes or elephant family groups), and stochastic formulations to capture extinction dynamics in small populations.

6. Conclusion

We developed and analyzed an SIRD+P compartmental model that couples epidemic dynamics with predator-prey interactions in a well-mixed prey population. Across seven ecological presets, We found that selective predation ($c \geq 2$) reduces epidemic severity as measured by peak infected count and total disease mortality, consistent with the healthy herd hypothesis. The magnitude of this benefit depends critically on the balance between predation intensity, prey birth rate, and predator population dynamics. Predation that is too intense relative to prey renewal destabilizes the population, making the prey more vulnerable to collapse than to disease. These results suggest that wildlife management interventions targeting predator populations should consider epidemic consequences alongside demographic ones, and that the epidemiological benefit of predators is contingent on their selectivity for diseased individuals.

GitHub Repository

All model code, preset configurations, and the interactive animated simulation are available at:

https://github.com/Apollo10101/CSCI5423_FinalProject

Interactive Simulation

An example video of the interactive animated simulation in action is available at:

<https://www.youtube.com/watch?v=J3cQmkxAPKc>

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